Multi-Agent Deal Review Pipeline: Architecture

Reviews a loan agreement against a compliance rule set and returns an evidence-backed report. LangGraph orchestration · Gemini · Pydantic contracts

Deterministic code · Gemini LLM · Hybrid

flow · retry · failure route

ORCHESTRATOR AGENT · LangGraph StateGraph manages shared state · routes tasks · enforces dependencies · retries · failure paths · audit log

citation failures → retry once with feedback

INPUTS

Deal document
PDF · TXT · MD

Compliance rule set
YAML: rules + escalation policy

1 Ingestion CODE
Splits the document into clause chunks with stable IDs (p2:7.2 = page 2, cl. 7.2). Loads + validates the YAML rule set. Fallback IDs p1:b2 for unnumbered docs.
writes: document, chunks, ruleset

2 Term Extraction GEMINI
Extracts ~28 material terms: parties, amounts, dates, rates, covenants, collateral... each with chunk_id, verbatim quote and confidence. Flags conflicts.
writes: extraction

3 Evidence Verifier CODE
Checks every quote really exists in the chunk it cites (normalized match). Hallucinated / unknown citations → feedback + one re-extraction.
writes: verified flags, feedback

4 Compliance Review HYBRID
Thresholds + derived LTV / tenor in code. Judgment rules: Gemini, one batched call. Grounding guard: PASS without valid evidence → NEEDS_REVIEW.
writes: compliance, derived_terms

compliance matrix

5 Risk & Summary GEMINI
Prioritized risk register (financial, legal, operational, compliance), gaps, executive summary, follow-up actions. Fallback: rule-based report from FAILs.
writes: risk_report

RiskReport

6 Human Escalation CODE
Applies the YAML escalation policy: high/critical FAILs, NEEDS_REVIEW, low-confidence or unverified terms, critical risks, agent errors.
writes: escalations

Escalation[]

7 Assemble Report CODE
Sets final status: completed / completed_with_errors / failed. Renders report.md + report.json, writes audit_log.json.
writes: status

on failure → skip to escalation (report still produced)

OUTPUTS

report.md · report.json
terms · matrix · risks · summary

audit_log.json
every agent step + LLM attempt

Streamlit UI · CLI
both call run_pipeline()

SHARED WORKFLOW STATE · PipelineState every node reads it and returns only the keys it changes · all values are Pydantic models

inputs	ingestion	extraction	compliance	risk	escalation	append-only
deal_path · rules_path	document · chunks · ruleset	extraction · verification_feedback	compliance · derived_terms	risk_report	escalations · status	errors[] · audit_log[]

Grounding & traceability

- Every term, rule result and risk cites p<page>:<clause>.
- Quotes are re-checked against the source in code.
- Invalid citations dropped; unsupported PASS downgraded.

Failure handling

- LLM output validated by schema; retry ≤2 with error feedback.
- 429: backoff; daily quota: fail fast. Agent down → fallback.
- Run ends completed_with_errors, never crashes; report always written.

Uncertainty & escalation

- Missing, unverified or low-confidence input never yields PASS.
- Conflicts (two effective dates) lower confidence + escalate.
- Thresholds live in the YAML policy, not in code.